Copernicus Global Land Operations "Cryosphere and Water"

"CGLOPS-2"

Framework Service Contract N° 945954 (JRC)

PRODUCT USER MANUAL

LAKE WATERS

100M PRODUCT

VERSION 2.1.0

Issue I1.02

draft – pending review

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List of Acronyms

ATBD	Algorithm Theoretical Basis Document
BC	Brockmann Consult
CCI	Climate Change Initiative
C-GLOPS	Copernicus Global Land Operations
FTP	File Transfer Protocol
GIO	GMES (Global Monitoring for Environment and Security) Initial Operations
GLWD	Global Lakes and Wetlands Database
GPT	Graph Processing Tool
LSR	Lake Surface Reflectance
NDWI	Normalised Difference Water Index
NERC	Natural Environment Research Council (UK)
obs	Observation
OC	Ocean Colour
OWT	Optical Water Type
PML	Plymouth Marine Laboratory
PUM	Product User Manual
QAA	Quasi-Analytical Approach
QAR	Quality Assessment Report
rep	representative
RMS	Root Mean Square
R_{rs}	Remote sensing reflectance
R_w	Fully normalised water-leaving reflectance
TS	Trophic State
TSM	Total Suspended Matter
TUR	Turbidity
WGS84	World Geodetic System 1984

1 BACKGROUND OF THE DOCUMENT

1.1 EXECUTIVE SUMMARY

The Copernicus Land Monitoring Service (CLMS) Lake Water Quality 100m (LWQ-100m) product provides an optical characterisation of inland waterbodies at 100m spatial resolution based on Sentinel-2 Multispectral Imager (MSI) data. This product was first provided between December 2018 and March 2020 (demonstration mode v1.3), to complement the LWQ-300m product that includes >4000 lakes globally and uses radiometrically and spectrally superior ocean colour instruments, but at lower spatial resolution. The 100m service has been continually active since July 2021 (v1.5) and has since been updated to the current product version (v2.0, v2.1), which aligns the 100m outputs with the 300m product. It continues to be provided in near-real-time.

Products are available for all resolvable water bodies within the outline of 225 Sentinel-2 MSI tiles over Europe and Africa. The 100m products contain six (sets) of variables: lake water-leaving reflectance (all wavebands available after atmospheric correction), turbidity, a trophic state index (derived from phytoplankton biomass by proxy of chlorophyll-*a*), total suspended matter concentration, chlorophyll-a concentration and a floating cyanobacteria risk index. Production and delivery cover up to 10-day intervals (starting the 1st, 11th and 21st day of each month), mapped onto a common global grid.

The current version (v2.1) is an update to the previous major release v2.0 (September 2024-October 2025). V2.1 contains a new scene classification layer based on the CLMS Water and Ice Cover (WIC) Sentinel-1+Sentinel-2 merged product. This new layer complements existing quality flagging in the LWQ-100m product, and provides an additional, independent source of lake ice/snow masking to users.

While Sentinel-2 MSI was primarily designed for land applications, some properties of waterbodies can also be observed, as demonstrated by the LWQ-100m product collection. The algorithms used to derive the input for the optical lake water products are implemented in the *Calimnos* processing chain, which also handles production of the Sentinel-3 OLCI LWQ-300m products. This Product User Manual (PUM) describes the product collection, including the latest developments in the current product version.

1.2 SCOPE AND OBJECTIVES

This document provides an overview on the Sentinel-2 MSI based lake water quality products at 100m (LWQ-100m) provided by the Copernicus Land Monitoring Service. The products follow the specification (CGLOPS2_SSD) of the Global Component of the Copernicus Land Service (CGLOPS), have been developed under a CGLOPS evolution contract and are currently delivered as demonstration products. The current product version is v2.1, which contains updates to scene classification compared to previous released versions. The products and information are freely available via the CLMS website (https://land.copernicus.eu/en/products/water-bodies?tab=lake_water_quality). For detailed information about the algorithms and quality

assessment, please refer to the Algorithm Theoretical Basis Document (ATBD) and Quality Assessment Report (QAR).

1.3 CONTENT OF THE DOCUMENT

This document is structured as follows:

- Chapter 2 recalls the user requirements and the expected performance
- Chapter 3 summarizes the retrieval methodology
- Chapter 4 describes the technical properties of the product
- Chapter 5 summarizes the results of the quality assessment including current limitations

1.4 RELATED DOCUMENTS

1.4.1 Applicable documents

AD1: Annex I – Technical Specifications JRC/IPR/2015/H.5/0026/OC to Contract Notice 2015/S 151-277962 of 7th August 2015

AD2: Appendix 1 – Copernicus Global land Component Product and Service Detailed Technical requirements to Technical Annex to Contract Notice 2015/S 151-277962 of 7th August 2015

AD3: Call for tenders JRC/IPR/2023/CPN/2325 – Framework Service Contract – Operation of the bio-geophysical variables systematic monitoring of the Global Land Component of the Copernicus Land Service 'CGLOPS' Part 2: Technical Specifications, Ref. Ares(2023)6699927, dated 03/10/2023

1.4.2 Input

Document ID	Descriptor
CGLOPS2_SSD	Service Specifications of the Global Component of the Copernicus Land Service
CGLOPS2_ATBD_LWQ100_S2_v2.1.0_I1.02	Description of algorithms for 100m lake water quality demonstration products
CGLOPS2_QAR_LWQ100_S2_v2.1.0_I1.00	Quality Assessment Report for 100m lake water quality demonstration products
CGLOPS2_ATBD_Polymer_v1.0.0_I1.02	Description of Polymer Algorithm for different instruments

2 REVIEW OF USER REQUIREMENTS

According to applicable document [AD3], the user requirements relevant for Lake Water Quality products are:

- **Definition**:

The Lake Water Quality products v2.1 are lake turbidity (TUR) in NTU, total suspended matter (TSM) in g/m^3, Trophic State Index (TSI), chlorophyll-a concentration (Chl-a) in g/m^3 and lake surface reflectance spectra (LSR, given in terms of fully normalized water-leaving reflectance R_w; a dimensionless quantity) from which TUR, TSM, Chl-a, TSI and the floating cyanobacteria index are derived. The products shall be provided as 10-day averages of the respective parameters, except for LSR for which the observation that is statistically most representative for the observed time period is given. The reason for the latter is that an average of LSR is not physically accurate and would not be useful to feed into alternative algorithms for the retrieval of optical water quality parameters. LSR describes the apparent colour of the water body. Turbidity and TSM are key indicators of water clarity. Turbidity quantifies the haziness of the water and acts as an indicator of underwater light availability. TSM quantifies the concentration of suspended particles in the water body. TSI, as defined by Carlson (1977), refers to the degree at which organic matter accumulates in the water body and is most commonly used in relation to eutrophication monitoring. Chl-a is the precursor product to Trophic State Index and is commonly used as a proxy to phytoplankton biomass. The cyanobacteria risk index indicates the detection of cyanobacteria surface blooms in the water body.

Geometric properties:

The baseline datasets described in this document shall be provided at a resolution of 100m. The target baseline location accuracy shall be 1/3 of the at-nadir instantaneous field of view. Pixel coordinates shall be given for the pixel centre.

- **Geographical coverage**:

Global window

The initial window definition is aligned to the global datasets produced during the GMES Initial Operations (GIO) phase for the most widely used output data:

- geographic projection: lat lon,
- geodetical datum: WGS84
- accuracy: min 10 digits
- coordinate position: pixel centre
- global window coordinates: UL: 180W-75N, BR: 180E, 56S

The following output specifications are further optimised with respect to the requirements:

- pixel size at 100m: 0.1/112°
- global window coordinates: UL: 180°W-90°N, BR: 180°E, 90°S
- global grid size at 100m: 400752 columns, 200376 lines

Continental windows:

Continental windows may be asked by the contracting authority to satisfy specific user needs. Wherever applicable, continental windows shall be drawn from global output data.

- **Accuracy requirements**:

Lake Water Reflectance is an Essential Climate Variable (GCOS 244, 2022) with an associated accuracy requirement threshold of 30% (breakthrough: 20% and target: 10%) at the reflectance peak (GCOS 245, 2022). Turbidity has an uncertainty threshold of 50% (goal: 35%) and trophic state has an uncertainty threshold of 0.3 in log10 or 50% [AD3].

- **Temporal Definition**

As a baseline, the biophysical parameters are computed by and representative of *dekads*, i.e. periods of nominally ten and varying from eight to eleven days, defined to start on days 1, 11 and 21 of each month of the year.

3 METHODOLOGY DESCRIPTION

3.1 OVERVIEW

CGLOPS Lake Water processing uses the *Calimnos* processing chain. *Calimnos* was initially developed in UK NERC GloboLakes (2012-2018) with input from the ESA Diversity-II project (2015) and has since been further developed with core collaboration between Plymouth Marine Laboratory (PML), University of Stirling, Brockmann Consult (BC) and HYGEOS in the Copernicus Land Monitoring Service, H2020 TAPAS, H2020 EOMORES, and ESA Lakes_cci activities. The chain can be used to process archived Envisat MERIS, Aqua-MODIS, Sentinel-3 OLCI and Sentinel-2 MSI data from archives or in near real-time. This document refers to 100m products based on MSI generated using *Calimnos v2.1*.

Calimnos combines into a single processing chain: data discovery, spatial resampling and subsetting by target area (individual water bodies, image granule or any other region of interest), radiometric and atmospheric corrections, pixel identification (land/cloud/water/ice), optical water type classification, individual algorithms (per parameter and water type), weighted blending of results, optional reprojection and spatiotemporal aggregation (Figure 1). For the production of 100m products *Calimnos* is operated on the Copernicus Data Space Ecosystem (CDSE).

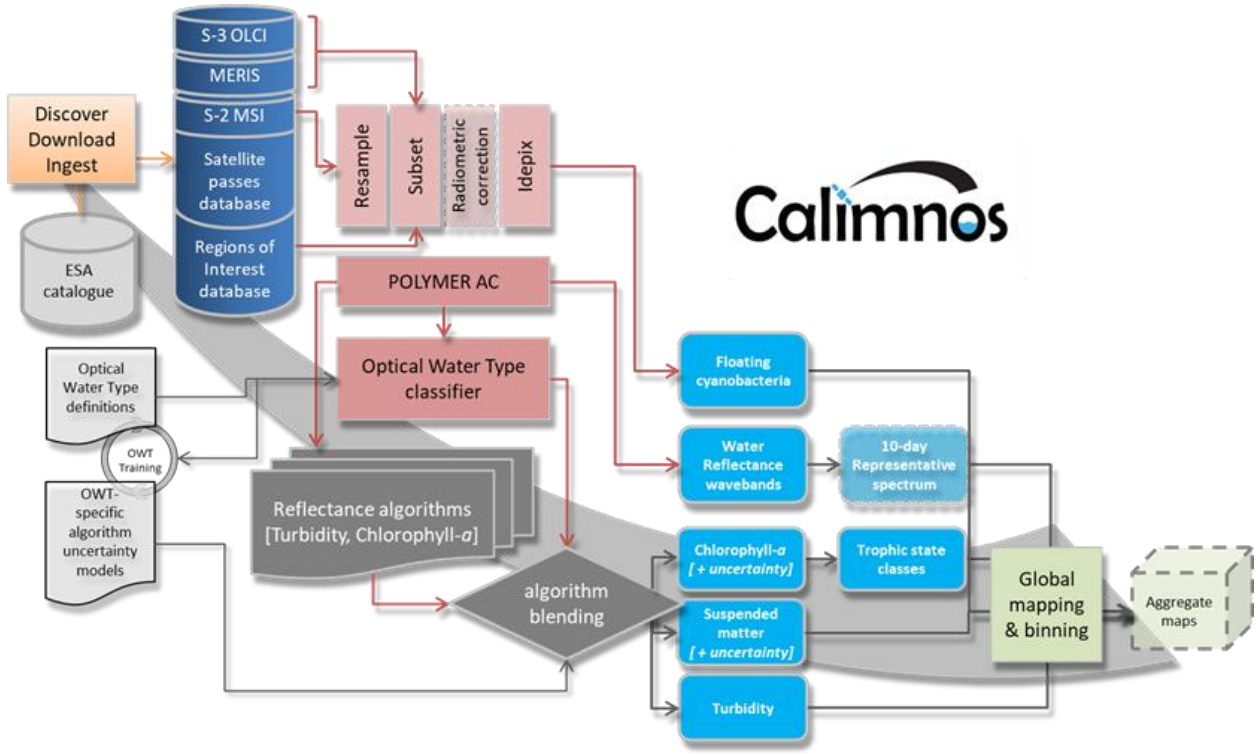


Figure 1: Schematic overview of the *Calimnos* processing chain v2.1, including MSI data processing

3.2 THE RETRIEVAL METHODOLGOY

3.2.1 Processing Chain and applied Tools

A schematic overview of *Calimnos* is given in Figure 1. The main processing stages include:

- Data discovery.
  - Granules matching regions-of-interest are identified and downloaded as they become available. Target regions are similarly specified in a geospatial database and satellite products which overlap any of the target regions can be queued for processing.
  - Tools: in-house Python scripts and PostgreSQL database functionality.
- Resampling.
  - The MSI bands at 10, 20, and 60m resolution are resampled to 100m. The Sentinel-2 resampling routine is part of the SNAP toolbox, called through the Graph Processing Tool (GPT).
  - Tools:
    - SNAP S2-resampling routine; retrieved resolution is 100m
    - SNAP Subset operator
- Subsetting.
  - This is an optional step and currently not applied.
  - Matching satellite passes can be subset to the area of interest. Currently, whole granules (UTM tiles) are processed.
  - Tools: The subsetting routine is part of the SNAP toolbox, called through GPT.
- Radiometric corrections.
  - This is an optional step which applies any radiometric corrections defined following the launch of the sensor to the L1C imagery before submitting the data to atmospheric correction. Lack of L1B (unmapped) imagery from Sentinel-2 is understood to prevent optimal image correction in the overlap between detectors, which cannot be corrected in this stage. Radiometric corrections are currently not applied to MSI.
- Pixel identification.
  - Pixel identification and classification over water with the *IdePix* algorithm for Sentinel-2. The algorithms combine several spectral tests and are applied for initial pixel identification as water, land, mixed pixels, cloud/haze, cloud shadow or snow/ice. Pixel identification masks are stored for later masking of invalid (non-water) pixels.
  - Tools: IdePix-Sentinel-2 v8.0.1 is called through SNAP using the GPT.
- Atmospheric correction.
  - Polymer is applied to the corrected L1C data that match the water mask derived from *IdePix* and yields water-leaving reflectance wavebands. Polymer is called using a function wrapper in Python.

- Tool: Polymer v4.14 according to Steinmetz et al. (2011), updated for Sentinel-2 application, parameterised to use the Park and Ruddick (2005) bidirectional reflectance distribution function, including mineral absorption and initialization of the optical model with chlorophyll-a = 10 mg m^{-3} and suspended matter = 10 g m^{-3} for case-2 waters, and operating only on pixels identified as water by the *IdePix* module.
- Polymer v4.15+ is the next candidate processor used for the Lake Water Quality products and includes a wider range of initialization conditions and range of backscattering albedo in the bio-optical model, compared to previous versions. The QAR document provides details on the performance of LWQ product v4.15 compared to v4.14. Product v2.0+ remains with the same atmospheric correction processor (Polymer v4.14), because downstream algorithms will need to be recalibrated before both can be included in the service, which is the current outlook towards LWQ-100m product v3.0.

- Optical water type classification.
  - Optical water type (OWT) classification derived from atmospherically corrected Sentinel-2 MSI imagery is applied to the image. Each pixel was initially assigned a membership score for four OWT classes. The classifier has been updated by including a wider range of optically diverse water bodies. With product version 2.0+, the OWT separates between 10 optical water classes.
  - Tool: Python at PML based on the work of Moore et al. (2001) and equivalent software developed for ESA Lakes_cci. The algorithm uses the spectral angle to determine similarity with predefined optical water types.
- Algorithm mapping and blending.
  - For each OWT, algorithms designed to be best suitable to describe chlorophyll-a and turbidity of that water type are blended on a pixel-by-pixel basis. A weighted average of turbidity and TSI (derived from Chl-a) is then produced based on the membership functions.
  - For OWTs where no algorithm performed well during validation no algorithm is assigned. Pixels with these OWTs are set to NaN.Algorithms CHL:
    - OWT class 1, 5: OC2 oceancolor.gsfc.nasa.gov/cms/atbd/chlor_a
    - OWT class 7: 708/665 empirical band ratio based on Gilerson et al. (2010)
    - OWT class 2,9: Semi-analytical NIR-Red band algorithm for MERIS based on Gons et al. 2005.
    - OWT class 3, 4, 6, 8, 10: unassigned.
  - Algorithms TUR: Nechad et al. (2010, 2016)
    - OWT class 1, 2: signal waveband 665nm
    - OWT class 5: signal waveband 705nm
    - OWT class 9: signal waveband 783nm
    - OWT class 7,10: signal waveband 865nm
    - OWT class 3, 4, 6, 8: unassigned

- Floating cyanobacteria / Macroalgae flag
    - Floating Algae Index (FAI) adopted from Dogliotti et al. 2018 and Oyama et al. 2015 to detect floating material and subsequent test based on Normalised Difference Water Index (NDWI) to distinguish between floating cyanobacteria and floating macroalgae.
- Temporal Aggregation
    - 10-day aggregation is done on a per-region basis after applying masks to select valid data for water pixels. The reflectance products of the Polymer atmospheric correction are inspected over the 10-day period and for each pixel the most representative spectrum is chosen (see Section 4). The aggregated products which contain the reflectance, chlorophyll concentration, trophic state index, total suspended matter and turbidity parameters are then reprojected to a global grid.
    - Tools:
        - SNAP Level-3 binner (LakeAggregate operator v1.3)
        - Most representative spectrum selection. Algorithm developed for CGLOPS between PML and BC. For each pixel, water-leaving reflectance spectra are compared to the median spectrum using the root-mean-square (RMS) difference across all wavebands. The spectrum with the lowest RMS difference is selected as most representative for the observation period.
- Mosaicking
    - The 10-day aggregated products are merged into a global grid using a Python script configured to select the pixel with the least spatial displacement in the region of overlap between MSI granules from the same orbit (Roy et al. 2016). This process is facilitated through a look-up table providing the pixel displacement for each pixel in the MSI UTM tiles compared to the 100-m global output grid, which is considered constant between orbits.

The SNAP version used is 8.8cv.

3.2.2 Basic underlying assumptions

The processing chain is based on a number of assumptions:

- Sentinel-2 MSI has sufficient radiometric sensitivity and dynamic range to capture water leaving reflectance in each band at 100m resolution, in dark, clear, and turbid waters.
- *IdePix* can differentiate adequately between pixels containing water and any of the following conditions: mixed land/water, cloud, cloud shadow, ice, snow, and haze.
- *Polymer* successfully retains the shape and amplitude of water-leaving reflectance.
- The optical water types derived from analysing MSI imagery over a diverse set of water bodies sufficiently capture the diversity of natural water types observed in CGLOPS.
- Algorithms calibrated to the narrow wavebands of MERIS and OLCI are applicable to the nearest located MSI wavebands, such that calibration of these algorithms yields results that are comparable with those from medium resolution imaging sensors.

- In case multiple algorithms for Chl-a/TSI and turbidity are used for different optical water types, it is assumed that the water-leaving reflectance spectra can be assigned a membership score for at least one known optical water type.
- The (tuned) reflectance algorithms for chlorophyll-*a* and suspended matter yield valid results over the range of reflectance values observed over water types to which they have been assigned.
- The translation of chlorophyll-*a* to trophic state is adequate and consistent between water types.
- Selection of the most representative spectrum from observations obtained over 10 days uses statistical metrics that result in a consistent pattern for the whole water body.

A crucial assumption in each version of the *Calimnos* processing chain is that predefined optical water types contain sufficient similarity to each satellite observation (pixel) containing open water. In practice, the assignment to an optical water type can fail due to a number of reasons, the most prominent being failure to identify the pixel as water (or as influenced by cloud or ice) by *IdePix*, or failure in the atmospheric correction leading to a low similarity to known optical water types. Optical water type membership scores are always assigned; poor similarity to any known type may result in downstream algorithms producing inconsistent results that require subsequent masking or flagging, which is done in case of extreme, or negative results.

3.3 LIMITATIONS OF THE PRODUCT

MSI, in contrast to ocean colour sensors, has relatively broad and fewer wavebands. It should be expected that, even when the bands are centred on diagnostic absorption features of optically active water constituents, sensitivity of the sensor to quantify these substances will be limited compared to ocean colour sensors. This can manifest in the products as having fewer valid outputs (more masked pixels), and result in showing a higher degree of uncertainty by comparison to coincident observations with ocean colour sensors.

The major limiting factor in (i) assessing the accuracy of water-leaving reflectance, (ii) the OWT-specific mapping of water constituent concentrations, and (iii) the accuracy of the water-column products is the scarcity of available in situ data to support algorithm tuning and validation. Without dedicated field campaigns, automated monitoring stations, and community data sharing arrangements, this remains a major source of product uncertainty. Insofar as OWT-specific algorithms have been validated for the MERIS and OLCI sensors, these can be used in the interim to conduct multi-sensor comparisons. Comparison of the Sentinel-2 derived products against better established algorithms for MERIS and OLCI is ongoing and will form the basis of re-tuned algorithms.

Some uncertainty remains regarding the treatment of observations within the overlap of adjacent MSI detectors during mapping to L1C. Without accurate unmapped (L1B) data, there is a risk that averaging of observations obtained under different viewing angles is not accurately handled at the atmospheric correction stage leading to visible camera boundaries in derived products.

The 100m products include near-shore areas that may be shallow. Such areas are not yet flagged. One method that was investigated (ESA Diversity-II project, 2015) consists of a long-term analysis to identify suspected shallow areas, but this method has not been globally verified.

3.4 DIFFERENCES WITH THE PREVIOUS VERSION

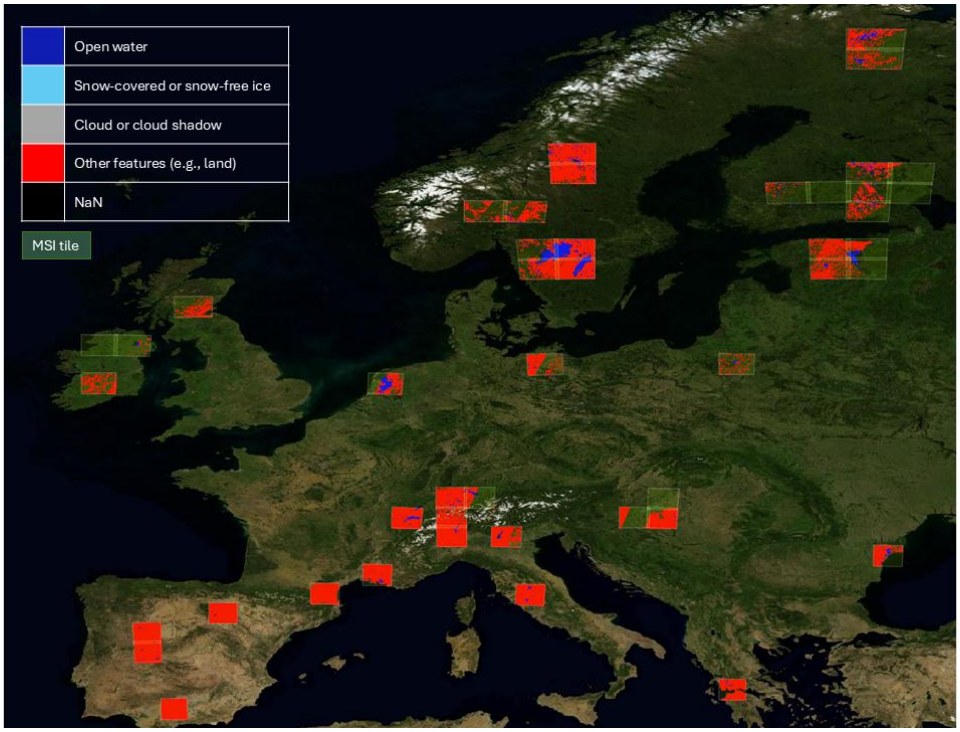
The LWQ-100m products have undergone several updates compared to the first version. The first version 1.3 (Dec 2018-March 2020) was a set of demonstration products generated for a set of 437 image tiles covering the largest lakes in Europe and Africa. By including whole granules, relatively smaller water bodies could be observed.

LWQ-100m v1.5 (July 2021- Nov 2024) covered 225 Sentinel-2 MSI tiles, which are a subset of the previous version. The selection of tiles has undergone consolidation by removing duplicate areas, tiles which mainly covered coastal waters, and several tiles in Scandinavia. Coastal areas are currently not yet masked out although the value of the product in these areas should be investigated.

LWQ-100m v2.0 introduced new bands to the product: chlorophyll concentration (input to TSI), total suspended matter, a floating cyanobacteria risk indicator, and per-pixel quality information. In v2.0, turbidity is no longer derived via suspended particulate matter but by direct calculation from water leaving reflectance bands.

LWQ-100m v2.1 (current version) differs from previous v2.0 in that it contains an new, additional scene classification layer that includes snow/ice pixel identification and classification information from the CLMS Water and Ice Cover (WIC) Sentinel-1 and Sentinel-2 merged product. This product uses Sentinel-1 synthetic aperture radar and Sentinel-2 MSI and is generated on a daily basis for inland water within the EEA38+UK area defined by a dynamic water mask that is updated annually. It provides information about river and lake areas covered by snow-covered or snow-free ice at 20m spatial resolution (HR-WSI PUM 2025). No WIC information is provided for MSI tiles outside the EEA38+UK region, as this is limited by the current geographical cover of the WIC product.

An example LWQ-100m v2.1 product (the ‘WIC_S1S2_aggregate’ layer) is shown below, overlaid on NASA’s Blue Marble map; first dekad in July 2025 (period 01/07/2025-10/07/2025):



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4 PRODUCT DESCRIPTION

4.1 FILE NAMING

For the 100m Lake Water Products, the naming convention will be as follows:

c_gls_<Acronym>_<YYYYMMDDHHmm>_<AREA>_<SENSOR>_<Version>.<EXTENSION>

where

- <Acronym> is the name of the product, which is **LWQ100** in the case of Lake Water Quality 100m.
- <YYYYMMDDHHmm> gives the temporal location of the file. YYYY, MM, DD denote the year, the month, the day of the product. The day is the first day of the dekadal average. Because time is not relevant due to temporal aggregation, HHmm is always set to 0000.
- <AREA> gives the spatial coverage of the file. In our case, <AREA> is either **GLOBAL** or the abbreviation of the continent.
- <SENSOR> gives the name of the sensor used to retrieve the product, for this version of the products it is **MSI**.
- The <Version> is currently **2.1**.
- <EXTENSION> is indicating the file format, which is **.nc** for netCDF4 files.

Example:

c_gls_LWQ100_202107010000_GLOBAL_MSI_V2.1.nc

4.2 FILE FORMAT

Sentinel-2 granules, optionally subset to regions of interest (water bodies) have been processed to NetCDF4 format compliant with CF convention 1.8.

4.3 COLLECTION IDENTIFIER

The collection **identifier** for version 2 of the LWQ-100m product is:

clms_global_lwq_100m_v2_10daily-nrt_netcdf

The **DOI citation** of the LWQ-100m product v2 collection is:

https://doi.org/10.2909/deae3db1-a214-4375-8d1a-63c42498050e

4.4 PRODUCT CONTENT

4.4.1 Data File

Each file includes the bands described in the table below.

Band name	Data type	Description
num_obs	int	Number of observations of the pixel area (including masked and invalid observations)
first_obs	float	Age in days after the start of the observation period of the first observation included in the pixel area
last_obs	float	Age in days after the start of the observation period of the last observation included in the pixel area
turbidity_mean	float	Mean of turbidity in Nephelometric Turbidity Units (NTU) (in the observation period)
tsm_mean	float	Mean of concentration of total suspended matter (in the observation period)
chla_mean	float	Mean concentration of chlorophyll-a (in the observation period)
trophic_state_index	float	Trophic state index (TSI) derived with *Calimnos* blending of chlorophyll-a algorithms tuned in the UK NERC GloboLakes project averaged over the observation period. The TSI has a value of 0, 10, 20, 30, 40, 50, 0, 70, 80, 90, or 100 corresponding to the classes given in Carlson (1977), where 0-40 are oligotrophic, 40-60 mesotrophic, 60-80 eutrophic, and 80-100 hypereutrophic.
R_w443_rep	float	Representative spectrum of fully normalised water-leaving reflectance at [the waveband indicated in the band name]
R_w490_rep	float	
R_w560_rep	float	
R_w665_rep	float	
R_w705_rep	float	
R_w740_rep	float	
R_w783_rep	float	
R_w842_rep	float	
R_w865_rep	float	
R_w945_rep	float	
R_w1375_rep	float	

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R_w1610_rep	float	
R_w2190_rep	float	
R_wDayNum_rep	float	Age in days after the start of the observation period of the most representative R_w spectrum
floating_cyanobacteria	float	Probability of floating cyanobacteria (with a range of values from 0 to 1 indicating probability of cyanobacteria presence – the closer the value is to 1, the higher the probability)
lat	float	Latitude in decimal degrees
lon	float	Longitude in decimal degrees
time	float	Time in days since 1-1-1970
crs	string	Geographical projection string
quality_flags	int	Flag band showing why input observations to a pixel have been included. Flags are L1_invalid, bright_pixels, cloud, snow_ice, land, turbidity_invalid, chla_invalid, tsm_invalid, atmospheric_correction_failure, floating_cyanobacteria, floating_macrophytes_or_mixed_pixel
WIC_S1S2_aggregate	int	Scene classification based on aggregated information from the CLMS Water and Ice Cover (WIC) Sentinel-1 and Sentinel-2 merged product (WIC S1+S2)

4.5 PRODUCT CHARACTERISTICS

4.5.1 Projection and Grid Information

The data are mapped to a global lat/lon grid at 0.1/112 degrees using WGS84 datum.

4.5.2 Spatial Information

The products currently cover 225 Sentinel-2 tiles at 100m resolution. The tiles cover water bodies in Europe and Africa, with emphasis on lake areas which are also available in the LWQ-300m service.

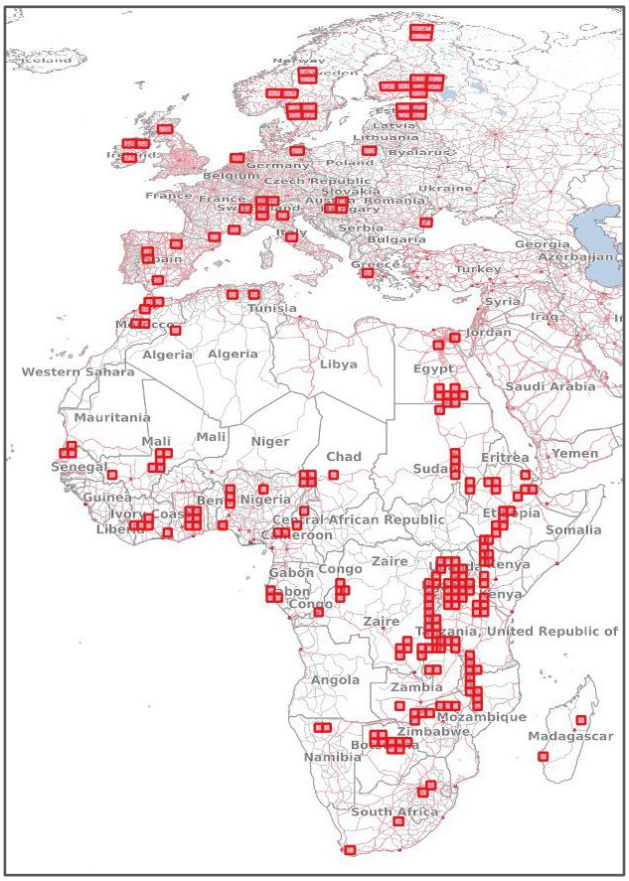
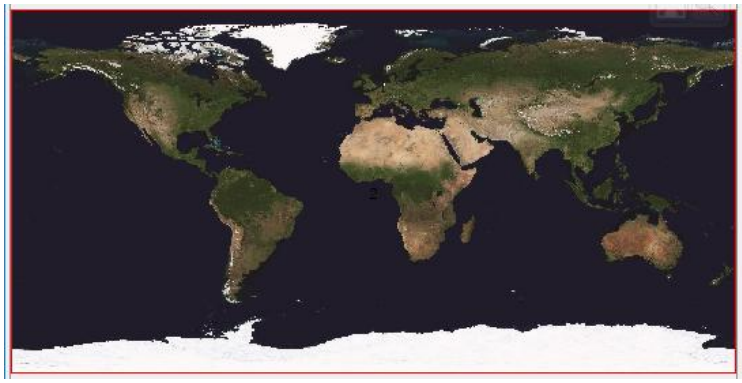


Figure 2: Distribution of 225 sentinel-2 tiles over Europe and Africa

All water pixels within those tiles are processed and provided. The tiles are mapped to the global raster.



Upper left latitude	89°59'55" N
Upper left longitude	179°59'55" W
Upper right latitude	89°59'55" N
Upper right longitude	179°59'55" E
Lower left latitude	89°59'55" S
Lower left longitude	179°59'55" W
Lower right latitude	89°59'55" S
Lower right longitude	179°59'55" E

Figure 3: Coverage of the global product (red rectangle) and bounding box coordinates

In the future, products may be provided in continental subsets following EUMETCAST subset definitions.

4.5.3 Temporal Information

The Lake Water Quality products are 10-day composites. The temporal information "YYYYMMDDHHmm" in the filename corresponds to the start date of the 10-daily period. The 10D averages comprise the following days of the month: 1-10, 11-20, 21-31. Consequently, the last "dekadal" period of the month varies in length between 8 and 11 observation days.

In addition to this (theoretical) observation time frame, the products contain the date of the (actual) first overpass contributing to the average and the (actual) last overpass. This information is stored in dedicated bands so that the observed period is known for each pixel (first_obs, last_obs). This information does not take into account whether a pixel is valid or not, but that a satellite observation was obtained.

4.5.4 Data Policies

Any use of the Lake Water Quality products implies the obligation to include in any publication or communication using these products the following citation:

"The products were extracted from the land service of Copernicus, the Earth Observation program of the European Commission. The Calimnos production chain was initially developed within UK NERC GloboLakes and has been further developed with core collaboration between PML, University of Stirling, Brockmann Consult and HYGEOS in the Copernicus Land Monitoring Service, H2020-TAPAS, H2020-EOMORES, and ESA Lakes_cci."

When using data or information from the service without modification:

"*Copernicus Service information [Year].*"

When using data or information from the service with modification:

"*Contains modified Copernicus Service information [Year].*"

Please send references of your publications using Copernicus Lake Water Quality products using our Service Desk https://land.copernicus.eu/en/contact-service-helpdesk.

For further details regarding the credits, copyright and legal ownership of the products, please refer to this Product User Manual and/or the relevant metadata, either in INSPIRE XML or inside the NetCDF data file.

4.5.5 Contacts

Accountable contact: European Commission Directorate – General Joint Research Centre

Email address: JRC-Copernicus-Land@ec.europa.eu

Scientific contact: BC and PML

Email address: lwq_info@brockmann-consult.de

Production and distribution contact: CLMS Helpdesk

https://land.copernicus.eu/en/contact-service-helpdesk

4.5.6 Sample products

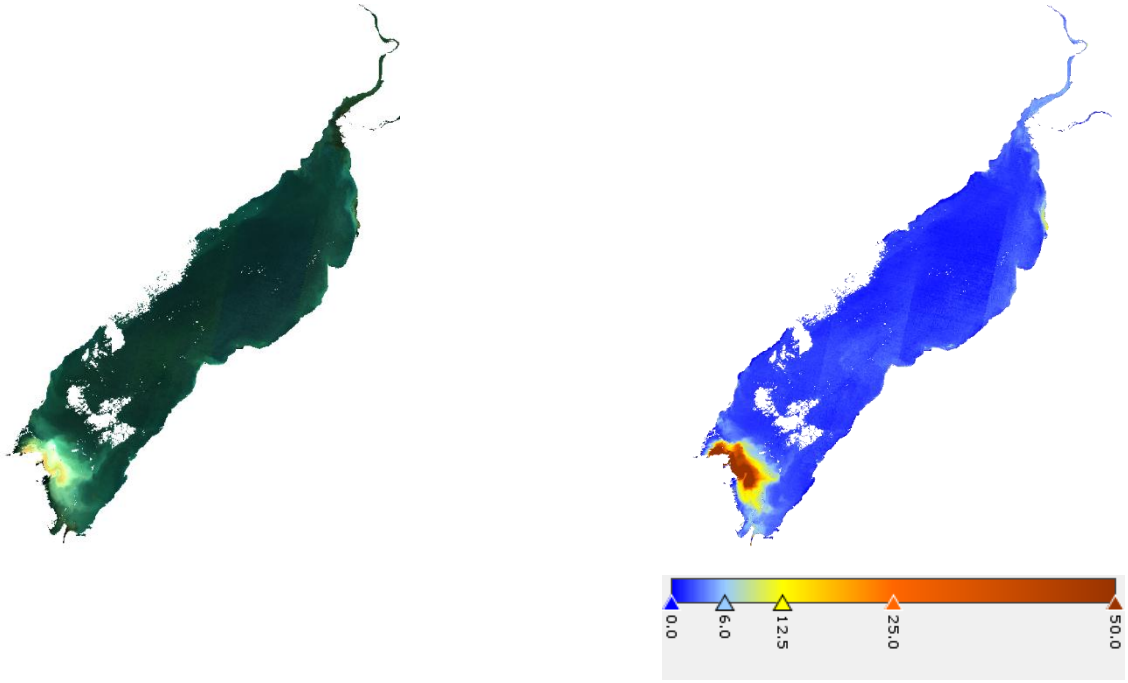


Figure 4: RGB derived from Lake Surface Reflectance at 665nm, 560nm, 490nm (left) and blended turbidity (right) for Lake Albert (Uganda, DR Kongo) in 100m resolution, first dekad of August 2019

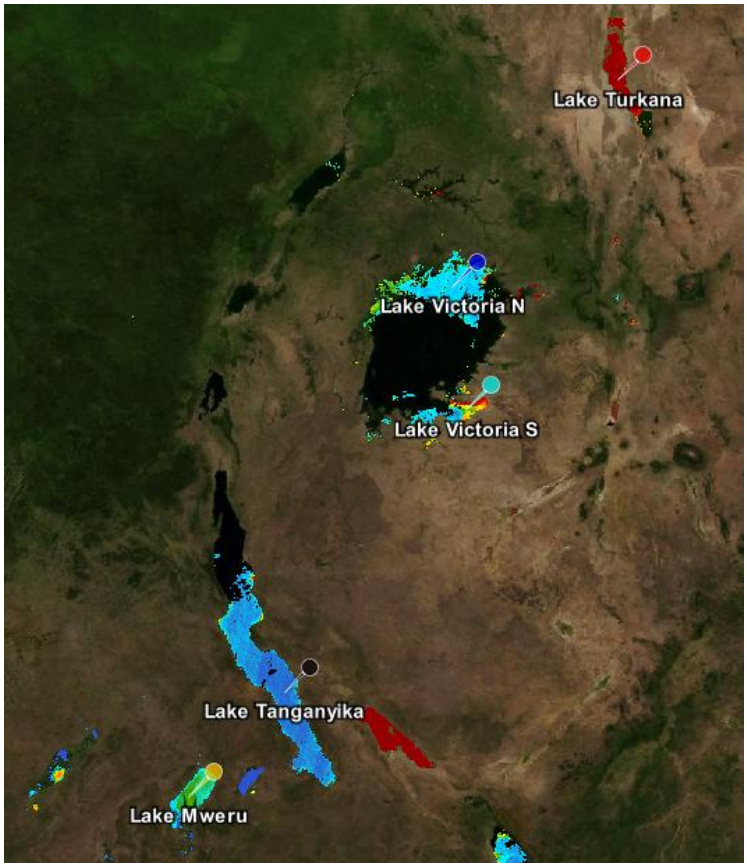


Figure 5: subset of 100m products showing Lakes in East Africa and locations for which spectra are shown in Figure 6

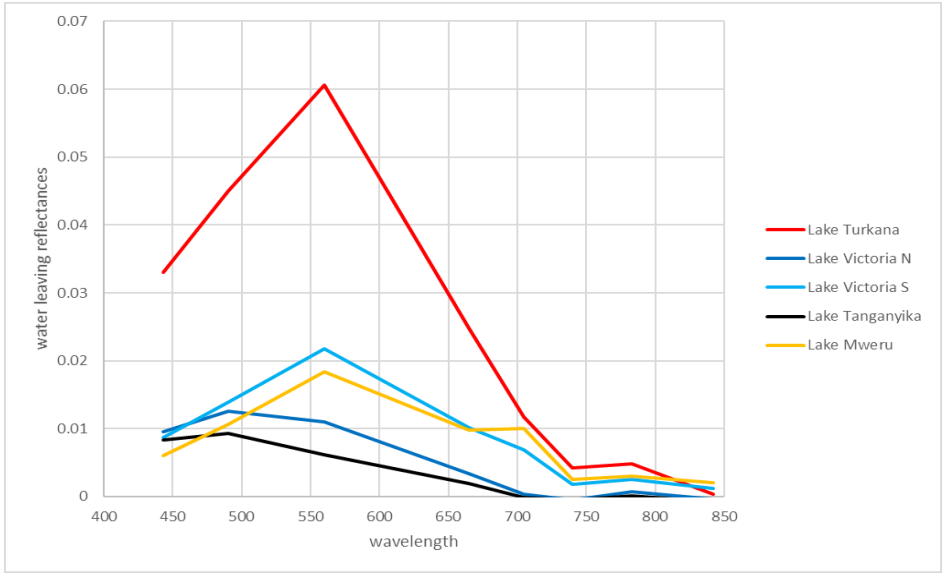


Figure 6: Water-leaving reflectance for selected lakes in East Africa, positions shown in Figure 6

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5 VALIDATION

This section is updated with validation results using the latest major product version update (v2.0). A detailed description of the validation and respective results from different product versions are provided in the QAR.

The performance of Polymer 4.14, as used in product versions v2.0 and v2.1, was assessed with comparisons against in situ R_{rs} from the GLORIA inland water reflectance dataset (Lehmann, 2023, Figure 7). R_{rs} for MSI from this Polymer version generally performs well in comparison with in situ R_{rs}. Lake Eire is underestimated when compared with the GLORIA dataset.

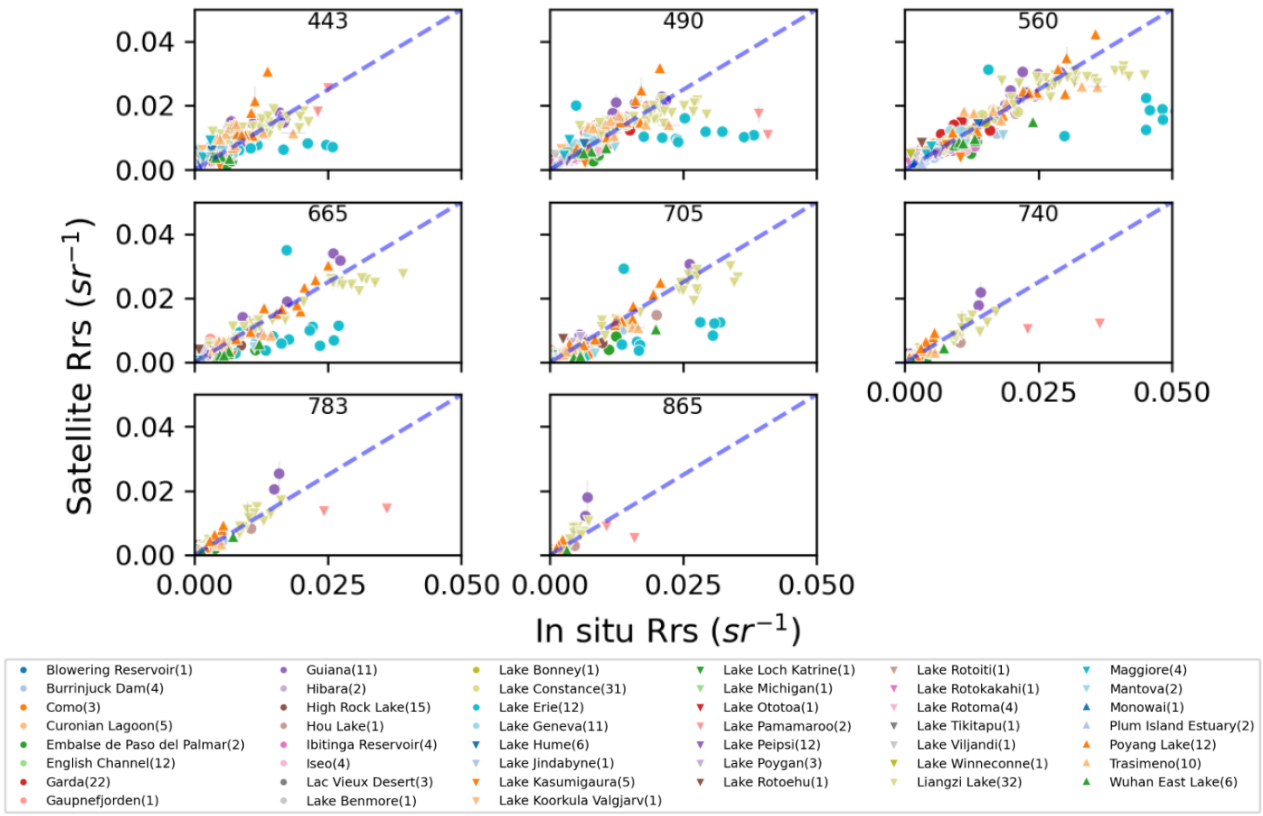


Figure 7: Matchups of in situ R_{rs} from the GLORIA inland water dataset and Sentinel-2 MSI R_{rs} derived using Polymer v4.14

Validation with in situ data remains an important task to do and will be continued, relying on the availability of such measurements.

6 REFERENCES

Bouvet, M., Ramoino, F. (2010). Equalization of MERIS L1b products from the 2nd reprocessing. ESA TN TEC-EEP/2009.521/MB.

Diversity-II (2015). ESA DUE DIVERSITY II Algorithm Theoretic Baseline Document (ATBD), Version 2.4, accessed here.

Dogliotti, A.I., Gossn, J.I., Vanhellemont, Q., Ruddick, K.G. (2018). Detecting and Quantifying a Massive Invasion of Floating Aquatic Plants in the Río de la Plata Turbid Waters Using High Spatial Resolution Ocean Color Imagery. Remote Sensing 10: 1140. https://doi.org/10.3390/rs10071140

GCOS 244 (2022). The 2022 GCOS Implementation Plan. https://library.wmo.int/records/item/58104-the-2022-gcos-implementation-plan-gcos-244. World Meteorological Organization. Accessed June 2024.

GCOS 245 (2022). The 2022 GCOS ECVs Requirements. World Meteorological Organization.

Gilerson, A.A., Gitelson, A.A., Zhou, J, Gurlin, D., Moses, W., Ioannou, I., Ahmed, S.A. (2010). Algorithms for remote estimation of chlorophyll-a in coastal and inland waters using red and near infrared bands. Optics Express 18(23): 24109-24125.

Gons, H.J., Rijkeboer, M., Ruddick, K.G. (2005). Effect of a waveband shift on chlorophyll retrieval from MERIS imagery of inland and coastal waters. Journal of Plankton Research 27(1): 125-7.

Lehmann, M.K., Gurlin, D., Pahlevan, N., et al. (2023). GLORIA - A globally representative hyperspectral in situ dataset for optical sensing of water quality. SciData 10: 100. https://doi.org/10.1038/s41597-023-01973-y

Moore, T.S., Campbell J.W., Feng, H. (2001). A fuzzy logic classification scheme for selecting and blending satellite ocean color algorithms. IEEE Transactions on Geoscience and Remote Sensing 39(8): 1764-1776.

Nechad, B., Dogliotti, A.I., Ruddick, K.G., Doxaran, D. (2016). Particulate Backscattering and suspended matter concentration retrieval from remote-sensed turbidity in various coastal and riverine turbid waters. Submitted for the Proceedings of ESA Living Planet Symposium, Prague, 9-13 May 2016, ESA-SP 740.

Nechad, B., Ruddick, K.G, Park, Y. (2010). Calibration and validation of a generic multisensor algorithm for mapping of total suspended matter in turbid waters. Remote Sensing of Environment 114: 854-866.

Nechad, B., Ruddick, K., Schroeder, T., Oubelkheir, K., Blondeau-Patissier, D., Cherukuru, N., Brando, V., Dekker, A., Clementson, L., Banks, A.C., et al. (2015). CoastColour Round Robin data sets: A database to evaluate the performance of algorithms for the retrieval of water quality parameters in coastal waters. Earth Syst. Sci. Data 7: 319-348.

Roy, D., Li, J., Zhang, H., Yan, L. (2016). Best practices for the reprojection and resampling of Sentinel-2 Multi Spectral Instrument Level 1C data. Remote Sensing Letters 7(11): 1023-1032.

World Meteorological Organization (2016). The Global Observing System for Climate: Implementation Needs. GCOS 200.

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